

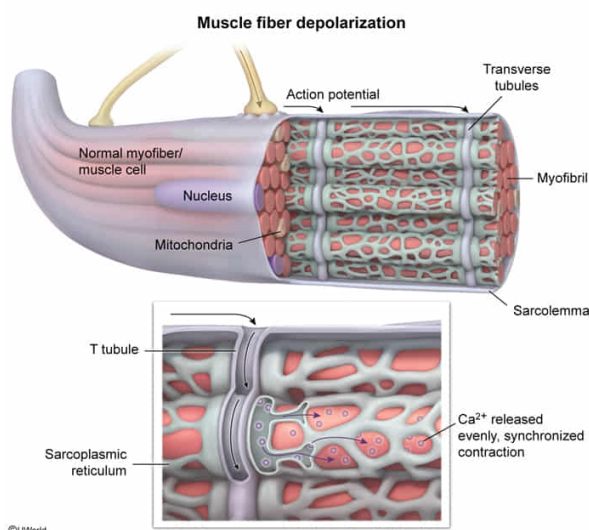
In neurons, action potential propagation occurs along the axon. The analogous structure in skeletal muscle fibers that enables action potential propagation is the:

- ☐ A. actin filament. [15%]
- ✓ ☒ B. transverse tubule. [68%]
- ☐ C. motor endplate. [7%]
- ☐ D. myelin sheath. [9%]

Correct

67% answered correctly

Explanation:



An action potential is a rapid change in the electric potential across a cellular membrane, where electric potential is the **difference in charge** between the intracellular and extracellular fluids. In the nervous system, the propagation of these electrical impulses from the presynaptic neuron can result in an action potential in the **postsynaptic neuron**; in the musculoskeletal system, action potential propagation through the muscle fiber leads to contractile activity.

Muscle fiber action potentials propagate along the **sarcolemma**, the plasma membrane of the muscle fiber. The sarcolemma burrows deep into the muscle fiber, forming hollow structures known as **transverse (T) tubules**. Action potentials propagate along T-tubules just as they propagate along the superficial sarcolemma, resulting in the **rapid** and **complete** depolarization of the muscle fiber. This promotes calcium release from the sarcoplasmic reticulum, leading to muscle fiber contraction.

(Choice A) Actin filaments are one component of the muscle fiber **contractile apparatus**. However, actin filaments are not involved in the transmission of action potentials.

(Choice C) The motor endplate refers to the muscle fiber component of the

neuromuscular junction (NMJ). The NMJ facilitates signal transmission between, but not within, cells.

(Choice D) The **myelin sheath** is a fatty layer of insulation that surrounds certain axons and increases the conduction speed of an action potential along the axon of the neuron. Although muscle fibers can be innervated at the NMJ, they do *not* contain neurons.

Educational objective:

Electrical depolarization caused by action potential propagation leads to the contraction of skeletal muscle fibers. As extensions of the plasma membrane that penetrate deep into muscle fibers, transverse (T) tubules are specialized muscle fiber structures that facilitate action potential propagation throughout individual muscle fibers.

Time spent:0 sec

Subject:
Biology

QID:401967

System:
Musculoskeletal System

The nicotinic acetylcholine receptor (nAChR) is a protein embedded in the muscle fiber membrane that facilitates cell-to-cell communication across the neuromuscular junction. When acetylcholine (ACh) binds nAChR, electrical depolarization within skeletal muscle fibers is induced and ultimately gives rise to contractile activity. The nAChR is composed of five protein subunits symmetrically organized around a central ion pore. Previous studies demonstrated that these subunits may serve as epitopes (antibody binding sites) in autoimmune disease. By reversibly binding nAChR subunits, anti-nAChR antibodies may temporarily block the attachment site for ACh, resulting in muscle weakness.

In preliminary studies for a research proposal, researchers harvested muscle samples from mice to investigate the effects of anti-nAChR antibodies on muscle tension, the pulling force transmitted by muscles during contraction.

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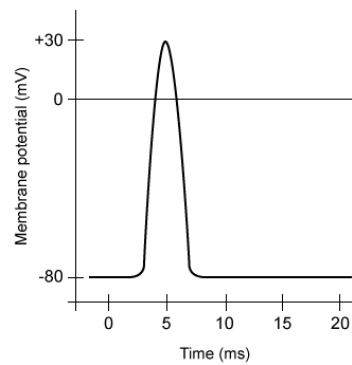


Figure 1 Action potential elicited from a harvested skeletal muscle sample

Experiment 2

The researchers divided the 30 muscle samples into three groups of 10. Resting muscle tension was measured, and the electrolyte-rich baths of each group were further infused with a 7.5-mL solution containing mouse-specific anti-nAChR antibodies, ACh, or a mixture of both (Table 1). Post-infusion (peak) muscle tension was measured for 30 minutes, with results displayed in Figure 2.

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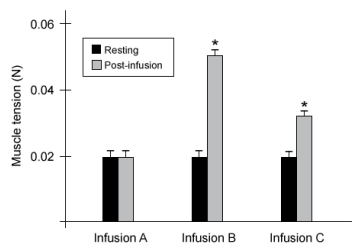


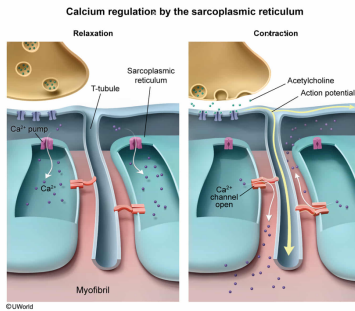
Figure 2 Pre-infusion resting muscle tension and post-infusion peak muscle tension in harvested muscle samples (Note: * indicates significantly different from all other groups)

Why was Ca^{2+} added to the oxygen-infused, electrolyte-rich bath? To allow Ca^{2+} to:

- ☒ A. be sequestered inside the sarcoplasmic reticulum of the muscle fibers. [67%]
- ☐ B. inhibit pyruvate entry into mitochondria of muscle fibers.
- ☐ C. increase the affinity of muscle fiber myoglobin for oxygen. [4%]
- ☐ D. facilitate the binding of ACh to its receptor on the muscle fiber. [27%]

Correct
67% answered correctly

Explanation:



Calcium (Ca^{2+}) is a mineral that plays an important role in the contraction of skeletal muscle fibers. The **sarcoplasmic reticulum (SR)** is a specialized endoplasmic reticulum in muscle fibers that is responsible for tightly regulating intracellular calcium concentration by **sequestering calcium ions** during periods of **relaxation** and **releasing them** into the cytosol during periods of **contraction** (in response to action potentials). Cytosolic calcium drives muscle contraction by first binding troponin, which then allows the actin and myosin filaments of the **sarcomere** to **slide** across one another.

The passage states that harvested muscle samples were placed in an oxygen-infused, electrolyte-rich bath that contained Ca^{2+} ions; it also states that these ions could enter the muscle fibers and enable maximal contractile force. Based on this information, the **calcium ions in the bath** were able to **enter** and **aggregate inside the SR** of the **muscle fiber** samples, enabling their contractile activity.

(Choice B) Muscle fibers can generate ATP through both **glucose** and **fat metabolism**. Pyruvate entry into mitochondria is necessary for **pyruvate oxidation** as well as subsequent use of pyruvate in the **citric acid cycle**, both of which occur inside mitochondria and are important to meet the energetic demands of muscle, particularly during contractions. Therefore, the researchers would *not* have added Ca^{2+} to *inhibit* pyruvate entry into mitochondria.

(Choice C) Iron, *not* calcium, facilitates oxygen binding by the proteins **myoglobin and hemoglobin**. Myoglobin stores oxygen within muscle fibers, and hemoglobin is an oxygen carrier molecule found in red blood cells that allows oxygen to be transported throughout the body.

(Choice D) At the neuromuscular junction, the entry of calcium ions into presynaptic neurons leads to the release of ACh from the presynaptic vesicles and into the synaptic cleft, where ACh then binds its receptor on the muscle fiber membrane. However, because the muscle fibers in the electrolyte bath were completely isolated, no presynaptic neurons were present, and calcium only served to supply the SR of the muscle fibers.

Educational objective:

The sarcoplasmic reticulum (SR) is a muscle fiber organelle that tightly regulates intracellular calcium (Ca^{2+}) concentration. The release of calcium from the SR promotes muscle fiber contraction whereas the transport of calcium into the SR promotes and maintains muscle fiber relaxation.

Time spent: 0 sec
Subject:
Biology

QID: 401968
System:
Musculoskeletal System

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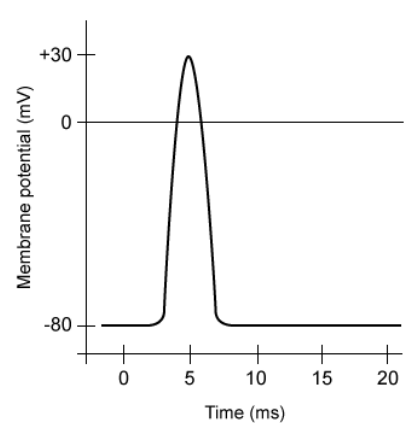


Figure 1 Action potential elicited from a harvested skeletal muscle sample

Experiment 2

The researchers divided the 30 muscle samples into three groups of 10. Resting muscle tension was measured, and the electrolyte-rich baths of each group were further infused with a 7.5-mL solution containing mouse-specific anti-nAChR antibodies, ACh, or a mixture of both (Table 1). Post-infusion (peak) muscle tension was measured for 30 minutes, with results displayed in Figure 2.

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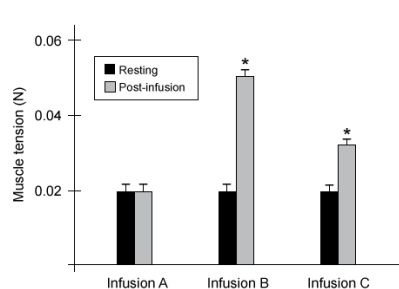


Figure 2 Pre-infusion resting muscle tension and post-infusion peak muscle tension in harvested muscle samples (Note: * indicates significantly different from all other groups)

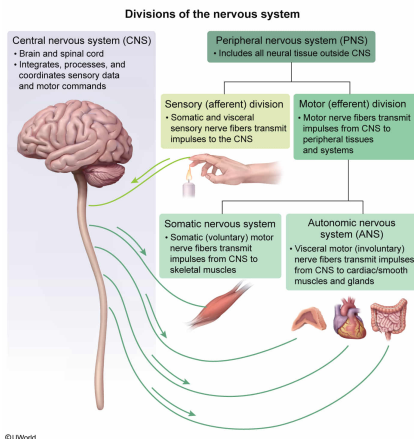
In humans, skeletal muscles are innervated by the somatic nervous system. An example of skeletal muscle function is mediation of:

- ☐ A. contraction of blood vessel walls. [3%]
☐ B. resting heart rate.
☐ C. propulsion of food contents along the digestive tract. [1%]
☒ D. movement of the jaw during chewing. [93%]

Correct

93% answered correctly

Explanation:



The **peripheral nervous system** consists of all nerves and neurons located outside the central nervous system (ie, the brain and spinal cord). The peripheral nervous system has two branches: the autonomic nervous system and the somatic nervous system.

The motor nerve fibers of the **somatic nervous system** transmit impulses from the central nervous system to the **skeletal muscles** of the body, allowing individuals to consciously control and perform specific movements. As a result, all movements carried out on a *voluntary* basis are mediated by the somatic nervous system. Examples of **voluntary movements** include activities such as walking, talking, and **movement of the jaw during chewing**.

In contrast, the **autonomic nervous system** mediates subconscious, automatic functions that are *not subject to voluntary control*. Accordingly, the activity of glands, smooth muscles, and cardiac muscles is regulated by the visceral motor nerve fibers of the autonomic nervous system. For example, the smooth muscles located within the walls of blood vessels can contract (**Choice A**) or relax to constrict or dilate the **vessel**, and the smooth muscle sheets in the walls of the gastrointestinal tract alternately contract and relax to propel food contents along the gut (**peristalsis**) (**Choice C**).

The contraction of cardiac muscle is another automatic, involuntary process regulated *in part* by the autonomic nervous system (**Choice B**). Although the action potentials necessary for cardiac contraction are generated by cardiac tissue independent of nervous system input, the activity of autonomic nerves influences the frequency and intensity of cardiac contraction.

Educational objective:

The somatic and autonomic nervous systems are the two branches of the peripheral nervous system. Voluntary activities occur through the contraction of skeletal muscles, which is mediated by the somatic nervous system.

Time spent: 0 sec

Subject:
Biology

QID: 401969

System:
Musculoskeletal System

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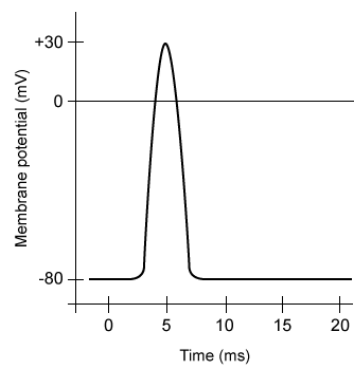


Figure 1 Action potential elicited from a harvested skeletal muscle sample

Experiment 2

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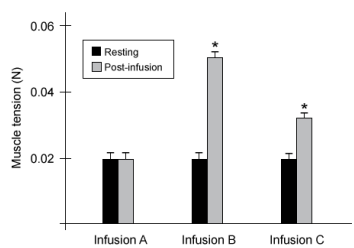


Figure 2 Pre-infusion resting muscle tension and post-infusion peak muscle tension in harvested muscle samples (Note: * indicates significantly different from all other groups)

A mutant mouse is found to exhibit nearly normal *in vivo* musculoskeletal function following the administration of anti-nAChR antibodies. Compared to a wild-type mouse administered the same antibodies, this mutant mouse is most likely to show which of the following changes at the neuromuscular junction?

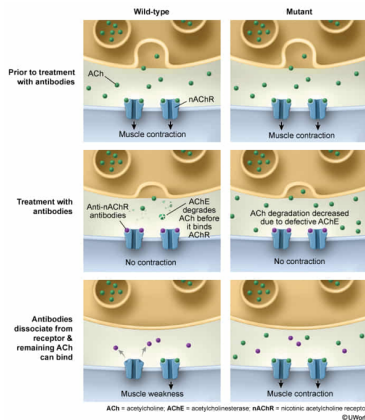
- ✔ ☒ A. Decreased ACh degradation within the synapse [58%]
- ☐ B. Increased expression of acetylcholine-degrading enzymes by skeletal muscle fibers [12%]
- ☐ C. Increased reuptake of ACh degradation byproducts into the presynaptic neuron [17%]

☐ D. Reduced release of ACh from the presynaptic vesicles [10%]

Correct

58% answered correctly

Explanation:



The interface between a somatic motor neuron and a skeletal muscle fiber is a specialized **synapse** known as the **neuromuscular junction (NMJ)**. The NMJ converts motor neuron action potentials into muscle fiber action potentials.

The arrival of an action potential at the axon terminal of the presynaptic motor neuron triggers release of ACh from presynaptic vesicles into the synaptic cleft via **exocytosis**. The ACh diffuses across the synaptic cleft to bind **nicotinic ACh receptors (nAChRs)** embedded within the membrane of the **postsynaptic muscle fiber**, eventually leading to muscle contraction. Unbound ACh may diffuse away from the synapse or be degraded by the enzyme **acetylcholinesterase (AChE)**, terminating muscle fiber contraction.

Based on the passage, anti-nAChR antibodies bind nAChR **reversibly** to block the interaction between ACh and nAChR, resulting in muscle weakness. Accordingly, administration of these antibodies in a wild-type mouse would temporarily prevent binding of ACh to its receptor. Consequently, the unbound ACh in the synaptic cleft of these wild-type mice would likely be **degraded** before being able to bind nAChR, decreasing muscle contraction.

However, if a mutant mouse were to exhibit normal muscle contraction despite being treated with anti-nAChR antibodies, a likely explanation could be that **ACh degradation within the mutant's synapse is decreased** compared to that of the wild-type. Decreased ACh degradation would **increase** the number of ACh molecules available to bind nAChRs after these receptors are no longer bound by anti-nAChR antibodies. This would likely allow muscles to eventually exhibit normal contraction despite the presence of anti-nAChR antibodies.

In contrast, the following changes would not likely mediate the improved muscle function observed in the mutant mice receiving anti-nAChR antibodies:

- Increased expression of ACh-degrading enzymes (ie, acetylcholinesterases) would decrease synaptic ACh, contributing to (*not* improving) muscle weakness (**Choice B**).
- ACh degradation products (eg, choline) can reenter the presynaptic terminal to be recycled for acetylcholine synthesis. However, increased reuptake of ACh degradation products would likely require *more rapid* ACh degradation to produce these recyclable materials. Consequently, increased ACh degradation would **decrease** ACh availability for nAChR binding, causing muscle weakness (**Choice C**).
- Following motor neuron action potentials, reduced release of ACh from the presynaptic vesicles would decrease the quantity of ACh released into the synapse, also leading to muscle weakness (**Choice D**).

Educational objective:

Acetylcholine (ACh) degradation within the neuromuscular junction contributes to termination of muscle contraction. However, decreased ACh degradation would allow more ACh molecules to remain in the synapse and prolong stimulation of nAChR and subsequent muscle contraction.

Subject:
Biology

System:
Musculoskeletal System

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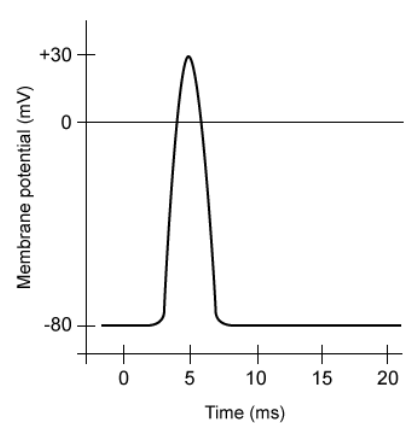


Figure 1 Action potential elicited from a harvested skeletal muscle sample

Experiment 2

The researchers divided the 30 muscle samples into three groups of 10. Resting muscle tension was measured, and the electrolyte-rich baths of each group were further infused with a 7.5-mL solution containing mouse-specific anti-nAChR antibodies, ACh, or a mixture of both (Table 1). Post-infusion (peak) muscle tension was measured for 30 minutes, with results displayed in Figure 2.

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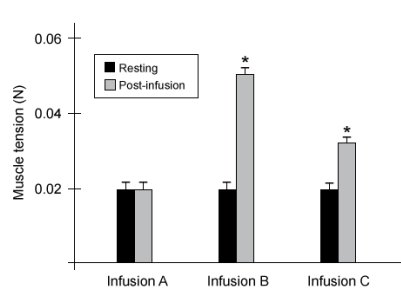


Figure 2 Pre-infusion resting muscle tension and post-infusion peak muscle tension in harvested muscle samples (Note: * indicates significantly different from all other groups)

To interpret the results of Experiment 2, researchers must assume that:

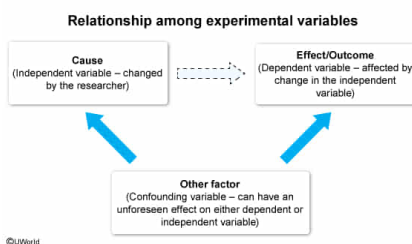
- I. muscle samples were of uniform weight and size.
- II. binding occurred between ACh and anti-nAChR antibodies in Infusion C.
- III. co-infusion of ACh and anti-nAChR antibodies was performed following ACh infusion.

- ✓ ☒ A. I only [43%]
☐ B. II only [4%]
☐ C. I and II only [44%]
☐ D. II and III only [7%]

Correct

43% answered correctly

Explanation:



The two primary types of variables in experiments are **dependent and independent variables**. The general goal of an experiment is to investigate the relationship between these two variables by intentionally modifying the independent variable and observing the effect of this modification on the dependent variable.

In Experiment 2, the researchers studied how three infusions, each containing different combinations of acetylcholine and anti-nAChR antibodies, affected the tension of muscle samples when added to an electrolyte-rich bath. The *infusion type* was intentionally modified and therefore was the *independent variable*. *Muscle tension*, which was assessed in each infusion, served as the *dependent variable*.

However, some experimental variables can have an unforeseen effect on the independent variable or dependent variable, complicating interpretation of a possible relationship between the two. Such variables are referred to as **confounding variables**. Ideally, confounding variables should be either **controlled** for or eliminated from an experimental design. Failure to do so may impair a researcher's ability to reach valid conclusions about experimental results.

For this scenario, if the muscle samples had varying weights and sizes, the muscle tension measurements with different infusions could not be compared in a valid manner because larger muscles would generate more tension than smaller muscles. Accordingly, to interpret the results from Experiment 2, the researchers must have assumed (or determined) that the muscle samples were of uniform (similar) weight and size (**Number I**).

(Number II) The passage states that the researchers were studying the effect of anti-nAChR antibodies on muscle tension. Specifically, these antibodies bind nAChR, disrupting its function and affecting muscle tension in such a way that can be measured by researchers. However, binding between ACh itself *and these antibodies* would *not* be required for interpretation of experimental results.

(Number III) Experiment 2 was conducted on three groups of 10 unique animal subjects. Isolated muscle samples from each group were exposed to the contents of *only one* infusion regimen (A, B, or C). At no point in time were these three groups treated with multiple infusion regimens.

Educational objective:

An ideal experiment features an independent variable, a dependent variable, and no confounding variables.

Time spent: 0 sec
Subject:
Biology

QID: 401971
System:
Musculoskeletal System

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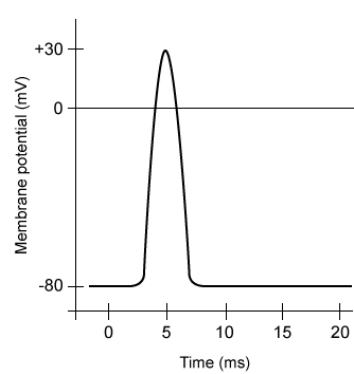


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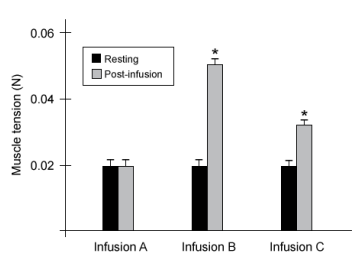


Figure 2 Pre-infusion resting muscle tension and post-infusion peak muscle tension in harvested muscle samples (Note: * indicates significantly different from all other groups)

The researchers proposed to use the same infusion protocol described in the passage to investigate how varying anti-nAChR concentration affected muscle tension in other mammalian animal models, but their research proposal was rejected by other scientists. Why was the proposal most likely rejected?

- ☐ A. Because resting muscle tension cannot be used as a control for post-infusion tension. [37%]
- ☒ B. Because the electrolyte-rich baths were infused with inconsistent amounts of ACh. [33%]
- ☐ C. Because skeletal muscles in the body would not be exposed to autoimmune antibodies. [22%]

☐ D. Because animal models cannot be used to study human diseases. [5%]

Correct

33% answered correctly

Explanation:

Proposed acetylcholine concentrations vary

	Infusion group	ACh (2.5 mg/mL concentration)	Anti-nAChR antibody (0.0 mg/mL concentration)
$7.5 \text{ mL} \times \frac{2.5 \text{ mg ACh}}{1 \text{ mL}} \rightarrow 18.75 \text{ mg ACh}$	A	0	7.5 mL
$3.75 \text{ mL} \times \frac{2.5 \text{ mg ACh}}{1 \text{ mL}} \rightarrow 9.36 \text{ mg ACh}$	B	7.5 mL	0
	C	3.75 mL	3.75 mL

The passage states that acetylcholine (ACh) binding to its nicotinic acetylcholine receptor (nAChR) induces muscle contraction, thereby causing muscle tension. Based on previous studies showing that nAChR subunits could be antibody binding sites, the researchers compared the effect of three different infusions on muscle tension: Infusion A (anti-nAChR antibodies only), Infusion B (ACh only), and Infusion C (ACh + anti-nAChR antibodies).

In this question, the proposed research objective was to investigate how varying concentrations of *anti-nAChR antibodies* affected muscle tension. Therefore, for the experiment described, the ACh concentration of infusions B and C should remain *unchanged*. This would allow researchers to assess how muscle tension was affected by varying concentrations of anti-nAChR antibodies *only*.

However, Table 1 shows that the ACh concentration of the infusions was **inconsistent**. Infusion B contained 7.5 mL of a 2.5 mg/mL ACh solution (ie, 18.75 mg of ACh). In contrast, **Infusion C** contained 3.75 mL of the same 2.5 mg/mL ACh solution (ie, **9.36 mg of ACh**). Consequently, proposed muscle tension measurements using the same protocol would be affected by varying concentrations of *both* ACh and anti-nAChR antibodies, instead of being affected by anti-nAChR antibodies alone.

(Choice A) Experiment 2 investigated the effect of various infusions on muscle tension. To quantify the positive or negative effect that each infusion has on muscle tension, a baseline condition (pre-infusion muscle tension, in this case) must be used as a control.

(Choice C) Like all **antibodies**, autoimmune antibodies (eg, anti-nAChR antibodies) are synthesized by **B cells** and transported through the blood stream to all body tissues (*including* skeletal muscle). Exposure of skeletal muscle to anti-nAChR antibodies would be a standard and expected characteristic of the autoimmune disease mentioned in the passage and would be unlikely to affect proposal approval or rejection.

(Choice D) Mice and other rodents are mammals that are frequently used as animal models to investigate mammalian physiology and pathophysiology (ie, diseases). Therefore, it is feasible that experimental results from mouse studies *can* be reproduced in other mammals (eg, humans).

Educational objective:

When the research goal is to assess how changes in a specific independent variable affect a dependent variable, researchers should ensure that any other variables that are not under study remain controlled or unchanged.

Time spent:0 sec

Subject:
Biology

QID:401972

System:
Musculoskeletal System